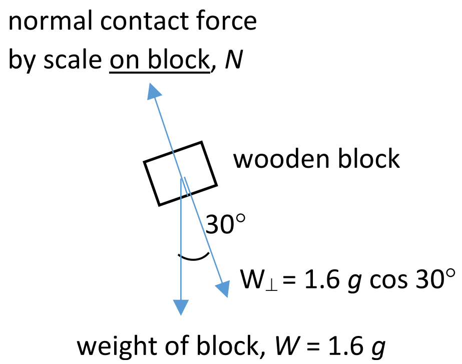
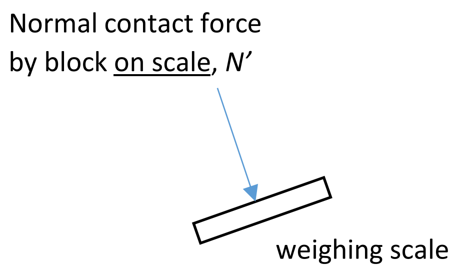
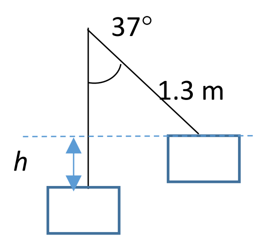
| Q1 | | Suggested Solutions | | Mark |
|---|---|---|---|---|
| 1 | (a) | <br><br>In the direction normal to the slope, the block is in equilibrium.<br>net force on the block = 0 N<br>thus, Normal contact force by the scale on the block, $N$<br>= weight component of the block normal to the slope<br>= 1.6 $g$ cos 30°<br>The scale measures the normal contact force exerted by the block on the scale, that is equal and opposite to the normal contact force by the scale on the block. (Newton's 3rd Law of Motion)<br>Thus, the reading on the scale<br>= normal contact force by the block on the scale/ $g$<br>= 1.6 $g$ cos 30° / $g$ = 1.39 or 1.4 kg. | | B1<br><br>B1<br><br>B1 |
| | (b) | (i) | Consider the block moving from the bottom to maximum height:<br>Let the velocity of the block immediately after collision be $v$.<br>By conservation of energy,<br>Loss in KE = Gain in GPE<br><br>$\frac{1}{2}mv^2 - 0 = mgh$<br>$v = \sqrt{2gh} = \sqrt{2(9.81)(1.3 - 1.3\cos 37°)} = 2.3 \text{ m s}^{-1}$ | B1<br>B1 |
| | | (ii) | applying N2L on the block: $\langle F \rangle = \frac{m(v-u)}{\Delta t} = \frac{1.6(2.3-0)}{0.2}$<br>$= 18.4 \text{ N}$<br>OR variations: $J = \langle F \rangle \Delta t = m(v-u)$ ; $\langle F \rangle = ma = m\frac{(v-u)}{\Delta t}$ | B1<br>A1 |

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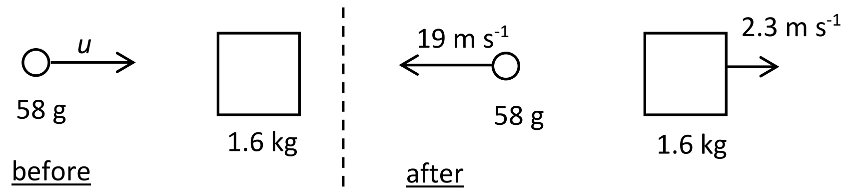
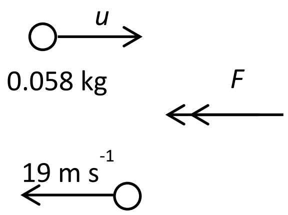
| | | **(iii)** | <br><br>By conservation of linear momentum, $m_{ball}u_{ball} = m_{block}v_{block} + m_{ball}v_{ball}$<br>$u_{ball} = (m_{block}v_{block} / m_{ball}) + v_{ball} = \frac{1.6(2.3)}{0.058} - 19 = 44.4 \text{ m s}^{-1}$<br>Or<br>$\langle F \rangle_{\text{on ball}} = \frac{m(v-u)}{\Delta t}$<br>$18.4 = \frac{0.058(19-u)}{0.2}$<br>$u = -44 \text{ m s}^{-1}$<br><br><br>i.e. v = 44.4 m s$^{-1}$ opposite to the final velocity. | **B1**<br><br>**A1** |
| | **(c)** | Relative speed of approach = 44 m s$^{-1}$<br>Relative speed of separation $= 2.3 + 19.1 = 21.4 \text{ m s}^{-1}$<br>$\neq$ relative speed of approach<br>Thus, the collision is not elastic.<br>OR<br>Initial KE = ½ (0.058)(44)$^2$ = 56 J<br>Final KE = ½ (0.058)(19)$^2$ + ½ (1.6)(2.3)$^2$ = 14.7 J<br>Since total KE s not conserved/ initial KE is greater than final KE, the collision is not elastic. | | **B1**<br><br>**A1** |

| Q2 | Suggested Solutions | Mark |
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